

Lecture Scribe Generation Instructions

L15_L17

 PDF

Purpose is to generate exam-oriented lecture notes (lecture scribe) strictly from the provided lecture PPT and my notes. The scribe must serve as a precise reference for exams [and exam preparation only].

Scope is to use only the content explicitly present in the provided material [which is the lecture PPT and relevant notes attached as context]. Do not add external explanations, background knowledge, interpretations or examples. If a definition, assumption, example, derivation, or proof is not present in the material, do not include it.

Structural constraint:

Follow the exact structure and order of the lecture PPT. Preserve topic sequence and hierarchy.

[Language and flow: Write in formal academic English. Keep sentences direct and precise. The tone should read like a student's own carefully written notes — not like an AI-generated document. Avoid unnecessary elaboration.]

Example Lecture Scribe (structure only)

Topic:

Sample Topic Title

Definition:

A formal definition as stated in the lecture.

Assumptions:

List of assumptions used.

Key Results:

Statement of a theorem or result.

Proof:

Step 1: First step as presented.

Step 2: Second step as presented.

Conclusion

Example:

A short worked example exactly as shown in the lecture.

Instruction

Using the same structure as the example above, generate the lecture scribe for the provided lecture PPT and notes. The scribe must:

Be structured exactly like the example

Include definitions, assumptions, results, proofs, derivations, and worked examples only if they appear in the provided material

Use simple, clear language while preserving correctness

Remain fully aligned with the lecture content and ordering

What a lecture scribe is?

A lecture scribe is a faithful, exam-ready reconstruction of what was taught in class, written so that a student can later rely on it as reference material while studying.

A correct lecture scribe should allow a student to:

Recall definitions and notation

Reconstruct proofs and derivations

Follow example solutions step by step

Understand logical dependencies between concepts

A lecture scribe should answer the question:

"If I study only this document, can I reliably revise what was taught?"

What it's not.

A lecture scribe is not:

A creative explanation

A simplified tutorial

A textbook rewrite

A solution manual for unseen problems

A place to add intuition not discussed in class

Everything included must be traceable to the provided context.



Topic:

Joint Random Variables and Joint Distributions

Definition:

- **Joint PMF (Discrete Case):** For discrete random variables $X \in \{x_1, x_2, \dots, x_n\}$ and $Y \in \{y_1, y_2, \dots, y_m\}$, the joint probability mass function is defined as $p(x_i, y_j) = P(X = x_i, Y = y_j)$. This computes the probability of the joint outcome occurring simultaneously.
- **Joint PDF (Continuous Case):** For continuous random variables X and Y , a function $f(x, y)$ is the joint probability density function if the probability that the pair (X, Y) lies in a small rectangle of area $dx dy$ is approximately $f(x, y)dx dy$.

- **Joint CDF:** The joint cumulative distribution function for random variables X and Y is defined as

$$F(x, y) = P(X \leq x, Y \leq y).$$

Assumptions:

- In the discrete case, the ordered pair (X, Y) takes values in the product set

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}.$$
- In the continuous case, if $X \in [a, b]$ and $Y \in [c, d]$, the pair (X, Y) takes values in the product region

$$[a, b] \times [c, d].$$
- When random variables are not independent, covariance and correlation are used as measures of the nature of the dependence between them.

Key Results:

- **Discrete Joint PMF Properties:**
 - $0 \leq p(x_i, y_j) \leq 1.$
 - $\sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) = 1.$
- **Continuous Joint PDF Properties:**
 - $f(x, y) \geq 0.$
 - $\int_c^d \int_a^b f(x, y) dx dy = 1.$
 - The joint PDF $f(x, y)$ may take values greater than 1; it is a probability density, not a probability.
- **Continuous Joint CDF Formulas:**
 - Obtained by integrating the density: $F(x, y) = \int_c^y \int_a^x f(u, v) du dv.$
 - Recovering the Joint PDF using partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y).$$
- **Discrete Joint CDF Formula:**

- Obtained by summing probabilities: $F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$.
- **Properties of the Joint CDF:**
 - $F(x, y)$ is non-decreasing; if x or y increase, then $F(x, y)$ must stay constant or increase.
 - $F(x, y) = 0$ at the lower-left of the joint range:
 $\lim_{(x,y) \rightarrow (-\infty, -\infty)} F(x, y) = 0$.
 - $F(x, y) = 1$ at the upper-right of the joint range:
 $\lim_{(x,y) \rightarrow (\infty, \infty)} F(x, y) = 1$.

Example:

- Example 1: Rolling Two Dice (Discrete PMF)
 - Experiment: Roll two fair dice.
 - Let X = value on the first die and T = total (sum) of both dice.
 - Each outcome (i, j) has a probability of $\frac{1}{36}$.
 - The resulting joint probability table consists of entries that are either 0 or $\frac{1}{36}$.
 - This establishes that X and T are not independent.
- Example 2: Joint PDF Validation and Calculation (Continuous PDF)
 - Given: $f(x, y) = 4xy$, for $0 \leq x \leq 1$ and $0 \leq y \leq 1$.
 - Step 1: Check Validity (Normalization).
 - $\int_0^1 \int_0^1 4xy \, dx \, dy = \int_0^1 (\int_0^1 4xy \, dx) dy = \int_0^1 2y \, dy = [y^2]_0^1 = 1$.
 - Because $f(x, y) \geq 0$ and the total probability equals 1, it is a valid joint PDF.

- Step 2: Compute $P(A)$ for event $A = \{X < 0.5, Y > 0.5\}$.
- $$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx = \int_0^{0.5} [2xy^2]_{0.5}^1 dx = \int_0^{0.5} 2x(1 - 0.25) dx = \int_0^{0.5} \frac{3}{2}x \, dx = \left[\frac{3}{16}\right].$$

Would you like me to generate a similar scribe for any other lecture materials you have?